



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
HALF YEARLY EXAMINATION

CLASS : X

SUBJECT :ROBOTICS & AI [SET A]

FULL MARKS :100

TIME : 2 HOURS

Instructions:

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this paper is the time allowed for writing the answers.

Attempt all questions from Section A and *any four* questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

This paper consists of **seven** printed pages.

SECTION A (40 Marks)
(Attempt all questions)

Question 1

- i) **Autonomous Mobile Robots (AMRs) navigate using:**
- | | |
|---------------------------|-------------------------------|
| a) Remote control | b) RFID only |
| c) Predefined tracks only | d) Cameras, sensors, and maps |
- ii) **Cobots are often used for small-batch production tasks because they:**
- | |
|--|
| a) Are faster than industrial robots |
| b) Require no programming |
| c) Are easily reprogrammable and re-deployable |
| d) Consume more power |
- iii) **Which company popularized the use of warehouse robots through its Kiva Systems?**
- | | | | |
|-----------|-----------|------------|----------|
| a) Google | b) Amazon | c) Walmart | d) Tesla |
|-----------|-----------|------------|----------|
- iv) **Infrared sensors are typically used in robots for:**
- | | |
|-------------------------------------|----------------------|
| a) Sound detection | b) Visual processing |
| c) Proximity and obstacle detection | d) Pressure sensing |
- v) **Which Pandas function returns the first 5 rows of a DataFrame?**
- | | | | |
|---------------|--------------|-------------|--------------|
| a) df.start() | b) df.head() | c) df.top() | d) df.peak() |
|---------------|--------------|-------------|--------------|

- vi) **What does the following code do?**

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3], [4, 5, 6])
plt.show()
```

a) Plots a histogram
b) Plots a bar chart
c) Plots a line chart
d) Raises a ValueError
- vii) **In a warehouse system using QR-code-based navigation for robots, the failure of which component would most likely halt operations?**
a) Ultrasonic sensor
b) Camera or vision sensor
c) Conveyor belt motor
d) Wireless antenna
- viii) **The primary reason helical gears are preferred over spur gears in high-speed applications is:**
a) They are cheaper to manufacture
b) They produce axial thrust
c) They have higher friction
d) They operate more quietly due to gradual engagement
- ix) **Assertion (A):** Passing the Turing Test proves that a machine is truly intelligent.
Reason (R): A machine can be considered intelligent only if it has self-awareness.
a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, but R is not the correct explanation of A.
c) A is true, but R is false.
d) A is false, but R is true.
- x) **What is the result of the following code?**

```
a = [1, 2, [3, 4]]
print(a[2][1])
```

a) 2
b) 3
c) 4
d) IndexError
- xi) **Assertion (A):** Gears are used to increase torque in machines.
Reason (R): Gears work by converting electrical energy into chemical energy.
a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, but R is not the correct explanation of A.
c) A is true, but R is false.
d) A is false, but R is true.
- xii) **What is the output of the following code?**

```
import pandas as pd
s = pd.Series([10, 20, 30], index=['a', 'b', 'c'])
print(s['b'])
```

a) 10
b) 20
c) 30
d) Error

- ### Question 2

- A/3

- iv) Explain how Cobots support human workers rather than replacing them. [2]
- v) Imagine you are managing a warehouse. Which type of robot would you use to improve efficiency and why? [2]

- vi) The following Python code reads a number and checks if it's a palindrome. But the code is not running, identify the problem and rewrite the code- [4]

```
num = input("Enter a number: ")
reverse = num[:0:-1]

if num = reverse:
    print("It is a palindrome number")
else
    print("It is not a palindrome number")
```

- vii) Read the following paragraph and answer the questions-

Cyber security is the practice of protecting computers, networks, programs, and data from unauthorized access, attacks, or damage. It ensures three main objectives: confidentiality, integrity, and availability of information. With the increasing use of digital platforms in every field, cyber security has become more important than ever. Common cyber threats include phishing, malware, ransomware, spyware, and denial of service (DoS) attacks. These threats can lead to data loss, financial damage, and privacy breaches. To safeguard against such risks, various security measures are used such as strong passwords, firewalls, antivirus software, encryption, and regular software updates. Cyber security plays a critical role in protecting individuals, businesses, and governments from cyber crimes, ensuring the safe and reliable use of digital technology. [4]

- a) What are the common cyber threats?
- b) What do you understand by cyber security?
- c) What measures we should take to safeguard ourselves?
- d) Why cyber security is important in today's world?

- viii) Consider the following code- [2]
- ```
L = [5, 10, 15, 20]
T = (25, 30, 35)
```

Write **one** statement each to:

- a) Replace the second element in L with 12
- b) Try to replace the second element in T and state what happens

## SECTION B (60 Marks)

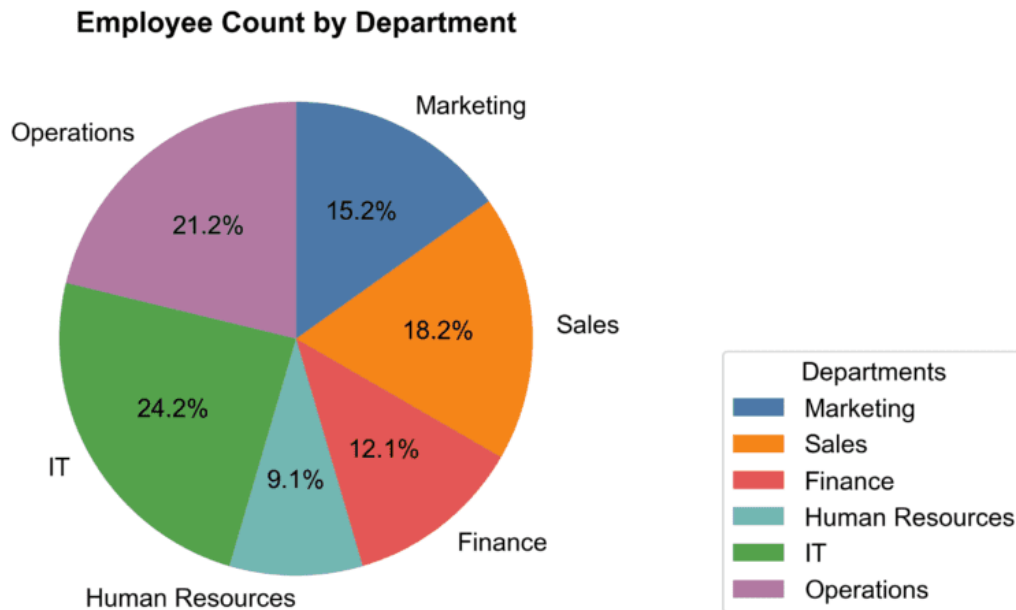
(Answer any four questions from this section)

The answers in this section should consists of the programs in either Python environment or any program environment with Python as the base. Each program should be written using variable description/ mnemonic codes so that the logic of the program is clearly depicted.

Flowcharts and algorithms are not required.

### Question 3

- a) Consider the following graph and write a Python program to construct it. [9]



- b) What is the difference between **addendum** and **dedendum** of a gear Explain with the help of an example. [3]
- c) What are autonomous robots? Name two such robots with their applications. [3]

### Question 4

- a) Write Python instructions for the following-
- Import NumPy, create an array of 5 integers and print it [2]
  - Create a 3x3 2D array to store first nine natural numbers [2]
  - Print the dimension of the above array [1]
  - Create two single dimension NumPy arrays and print their sum and product [2]
  - Print the shape and size of the above array [2]
- b) What do you understand by **gear ratio**? What is it used for? Consider a gear system consist of two spur gears; a driving gear with 30 teeth and a driven gear with 90 teeth. Calculate the gear ratio. [3]
- c) List out **three** points where Artificial Intelligence has an edge over Human Intelligence. [3]

### Question 5

- a) Write **Python** statement to do the following-
- i) Create a tuple of five integers and print it [1]
  - ii) Access the third element of the tuple [1]
  - iii) Slice the tuple to get the first three elements [1]
  - iv) Check if number 12 exists in the tuple [1]
  - v) Print the length of the tuple [1]
  - vi) Create another tuple of integers and concatenate it with the above made tuple. [1]
  - vii) Find the highest and the lowest value of the first tuple. [1]
  - viii) Use a for loop to print the items of the first tuple [2]
- b) Discuss any three types of sensors with examples. [3]
- c) What is deep **fake AI**? How it is impacting the society? What is your strategy to combat the situation? [3]

### Question 6

- a) Write a **Python** statement to do the following-
- i) Create a list of five flowers [1]
  - ii) Add a new flower at the end of the list [1]
  - iii) Add a flower 'Tulip' in the third position of the list [1]
  - iv) Delete 'Tulip' from the list [1]
  - v) Check if 'Rose' exist in the list [1]
  - vi) Remove all the duplicate entries from the list [1]
  - vii) Arrange the list in descending order [1]
  - viii) Print the number of occurrences of 'Lily' [1]
  - ix) Convert the list into tuple [1]
- b) What are the challenges associated with sensor integration in Robotics? Discuss any three briefly. [3]
- c) Describe how **Turing Test** works? List out the different variations of Turing Test?[3]

### Question 7

- a) Write the following code in **Python**-
- i)

|   | Name    | Maths | Science |
|---|---------|-------|---------|
| 0 | Martha  | 87    | 83      |
| 1 | Tim     | 91    | 99      |
| 2 | Rob     | 97    | 84      |
| 3 | Georgia | 95    | 76      |

- ii) Create the above **Data Frame** and display its content. [4]
- iii) Create a **Series** of four integers such as 87,89,91,90. Change their index to A,B,C,D. Display the **Series**. [2]
- iv) Add the above **Series** to the **Data Frame** under heading 'AI' [2]
- v) Print the **Data Frame** [1]
- b) What do you understand by Pitch Circle, Base Circle and Root Circle of a **Gear**? [3]
- c) How NumPy arrays are different from List [3]

### Question 8

- a) Write a Python program to read the user's name and number of calls made. Calculate the **total bill amount** to be payable based on the following slab- [9]

| <u>Calls made</u>   | <u>Rate</u>       |
|---------------------|-------------------|
| Upto 100 calls      | no charge         |
| Next 100 calls      | 90 paise per call |
| Next 200 calls      | 80 paise per call |
| More than 200 calls | 70 paise per call |

However, every customer has to pay Rs. 180 as monthly rent for availing the services.

- b) What do you understand by a **smart city**? List out a few features that makes a city smart. [3]
- c) Humans are said to be Intelligent, list out the factors that makes us Intelligent. [3]